

CS 8803/4803 Sociotechnical Security

Semester Project Proposal Guidelines

Georgia Tech, Fall 2026

Introduction

Over the course of the semester, you will work in groups of 1–3 people to conduct original research in Sociotechnical Security. This document details what you will need to submit for your project proposal. **If you are working as part of a team, each team member needs to upload a copy of the proposal on Gradescope**

Your 1-2 page proposal should outline your project's initial direction and is worth 10% of your grade. It will be evaluated on clarity, relevance to course themes, feasibility (i.e., can you achieve your goals in the remainder of the semester?), and methodological soundness (e.g., is it clear what you will accomplish).

You should use either the USENIX Security Overleaf template ([Overleaf v.2023](#)) or the Word template ([Word v.2022](#)) for your proposal. This is the same template as the mid-point check-in document (so feel free to continue using the same document).

Below are the items you must include in your submission.

Team member names

Please include your name, program/year, and email address at the top of your submission. If you are working in a group of 2-3, please include all of your team members' names, program/year, and email address (including your own), at the top of your submission.

Project options

Please indicate somewhere on your document which of the two project options you are completing. Remember: The expectation of effort for both options is the same – the latter simply provides more structure while the former is more open-ended. The project milestones, submission requirements, and grading criteria are the same for both options.

- Option One: Original Research Idea (Develop your own project idea related to Sociotechnical Security and execute it)
- Option Two: Platform Measurement Project (Select a platform and harm type, develop an automated detector, and analyze the system.)

Working title

A concise, descriptive title for your project (this can evolve later).

Project summary

Describe the main focus of your proposed group project and why it matters; this is a little bit like a pitch. A good summary will set the scene as to the problem or harm you are going to be looking at for the remainder of the semester. Setting the scene means explaining (briefly) what research has been done before, or citing a mainstream media article which highlights an issue.

You do not need to include a comprehensive survey of related work in the proposal, but you should cite a handful of articles that motivate your project and you should have a high-level sense of what prior work has already done so that you are not pitching an idea that has already been worked on.

Finally, a good summary gives a general overview as to what you intend to do, including the methods, the type of data you will be analyzing, and the expected findings. No need to go into detail on the methods here; you will be describing methods in depth in a later part of your proposal.

Research questions and motivation

Following your project summary, you should explain what research question(s) you are going to be investigating, and specifically what motivates this focus; what is missing or overlooked from previous approaches? This can be through appeals to who or what might be impacted by this harm.

Please make sure that by this point in your proposal, it is clear to a reader how your project relates to course themes.

Research or measurement plan

Outline your intended method or approach, depending on your project type. Being clear as to what type of method you choose is important as you will be evaluated on how feasible it is that you can accomplish the project using it. If are going to rely on human participants (e.g., via interviews, surveys, cognitive walkthroughs), you must make sure you'll be able to successfully recruit enough participants. This section should outline the types of data you are looking to collect (e.g., forum content, metadata), how much of it, and the type of analysis you will be applying.

You should be able to answer the following questions in this section:

Option One: Original Research Idea

1. What are your guiding research questions
2. What methods do you plan to use (e.g., interviews, content analysis, audits, simulation, system walkthroughs, web crawling, etc)
3. What is your anticipated data source or site of inquiry?
4. What will you be measuring, mapping, or evaluating?

Option two: Platform Measurement Project

1. What are your guiding research questions?
2. What harm type will you study (e.g., harassment, deception, sexualization, exploitation, etc)?
3. What platform or ecosystem will you focus on?
4. What automated detection approach (keyword matching, classifier, rule-based, etc) might you build?
5. How will you test your tool for accuracy?

Institutional Review Board for human-subjects research

Optional: If you plan to collect data from human participants and would like the option of continuing to work on your project after the course (e.g., to submit for publication in an academic conference), or if you plan to collect data from minors, you must submit an Institutional Review Board (IRB) request prior to the collection of any data. Veronica can help with this if needed. **If you anticipate wanting to do this, please indicate in your proposal that you will be submitting to the IRB.** If you do not plan to continue working on your project after the course, or you will not be collecting data from minors, you do not need to submit an IRB.

Anticipated challenges and risks

Demonstrate foresight and critical awareness of the ethical, technical, and practical risks inherent in your proposed project.

A strong proposal will describe not just the challenges you expect to face in collecting your data, but also how you will work around those. For example, if you can't recruit participants from a particular community, how will you quickly pivot your project in a way that will make it successful?

Beyond practical risks, you must also reflect on the ethical implications of the work that you expect to do. Describe how and why specific risks may emerge from your research activities, and what practical steps you will take to mitigate them.

This section will be evaluated based on specificity, realism (i.e., will something actually happen), and fit between risks and suggested mitigations. The best submissions will include ethical reasoning with technical awareness and show that you have seriously considered feasibility, safety, and integrity in your research proposal.

Expected outcomes and how you will measure success

Explain what your team will hope to deliver by the end of the semester. In other words, if you succeed, what will you contribute to the sociotechnical security research community that others could learn from and build on? This could be anything from a prototype of a detector of cyberbullying on Snapchat, to a deep dive into how policy approaches to moderating pro-anorexic content has adapted over time across three specific platforms, or offer a series of a design recommendations for child-safe smartphones.

Please also include a brief statement of how you will know that you have achieved success (e.g., that your project “worked”)? This should be more than “I finished the project”.

Division of Roles (Optional)

If you are working as part of a team and if you have a plan for how your group will divide work across members (e.g., data collection, analysis, writing, visualization, communication, etc), please include it in your proposal. This section will not be graded. However, this is a helpful exercise for thinking through who will be doing what and areas where you might have shared responsibilities.